

Newman-Penrose Formalism Calculations

Stephani et al., Equation (22.17)

Metric

$$ds^2 = -f dt \otimes dt + \rho dt \otimes d\phi + \rho d\phi \otimes dt + \frac{1}{\sqrt{\rho}} d\rho \otimes d\rho + \frac{1}{\sqrt{\rho}} dz \otimes dz$$

Coordinates

$$(x^\mu) = (t, \phi, \rho, z)$$

Metric Functions

$$f = 4b^2\rho^2 + c\rho\log(\rho)$$

Null Tetrad

Covariant components

$$l = (1) dt$$

$$n = \left(\frac{1}{2}f\right) dt + (-\rho) d\phi$$

$$m = \left(\frac{\sqrt{2}}{2\rho^{\frac{1}{4}}}\right) d\rho + \left(\frac{i\sqrt{2}}{2\rho^{\frac{1}{4}}}\right) dz$$

$$\bar{m} = \left(\frac{\sqrt{2}}{2\rho^{\frac{1}{4}}}\right) d\rho + \left(-\frac{i\sqrt{2}}{2\rho^{\frac{1}{4}}}\right) dz$$

Contravariant components

$$l = \left(\frac{1}{\rho} \right) d\phi$$

$$n = (-1) dt + \left(-\frac{f}{2\rho} \right) d\phi$$

$$m = \left(\frac{1}{2} \sqrt{2} \rho^{\frac{1}{4}} \right) d\rho + \left(\frac{1}{2} i \sqrt{2} \rho^{\frac{1}{4}} \right) dz$$

$$\bar{m} = \left(\frac{1}{2} \sqrt{2} \rho^{\frac{1}{4}} \right) d\rho + \left(-\frac{1}{2} i \sqrt{2} \rho^{\frac{1}{4}} \right) dz$$

Directional Derivatives

$$D = \frac{1}{\rho} \partial_\phi$$

$$\Delta = -\partial_t - \frac{f}{2\rho} \partial_\phi$$

$$\delta = \frac{1}{2} \sqrt{2} \rho^{\frac{1}{4}} \partial_\rho + \frac{1}{2} i \sqrt{2} \rho^{\frac{1}{4}} \partial_z$$

$$\bar{\delta} = \frac{1}{2} \sqrt{2} \rho^{\frac{1}{4}} \partial_\rho - \frac{1}{2} i \sqrt{2} \rho^{\frac{1}{4}} \partial_z$$

Spin Coefficients

$$\rho = 0$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = 0$$

$$\lambda = 0$$

$$\nu = \frac{1}{4} \sqrt{2} \rho^{\frac{1}{4}} \frac{\partial}{\partial \rho} f - \frac{\sqrt{2} f}{4 \rho^{\frac{3}{4}}}$$

$$\alpha = -\frac{\sqrt{2}}{16 \rho^{\frac{3}{4}}}$$

$$\beta = -\frac{3 \sqrt{2}}{16 \rho^{\frac{3}{4}}}$$

$$\tau = -\frac{\sqrt{2}}{4 \rho^{\frac{3}{4}}}$$

$$\pi = \frac{\sqrt{2}}{4 \rho^{\frac{3}{4}}}$$

$$\epsilon = 0$$

$$\gamma = 0$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = 0$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = \frac{1}{4} \sqrt{\rho} \frac{\partial^2}{(\partial \rho)^2} f - \frac{\frac{\partial}{\partial \rho} f}{4 \sqrt{\rho}} + \frac{f}{4 \rho^{\frac{3}{2}}}$$

$$\Lambda = 0$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = \frac{1}{8 \rho^{\frac{3}{2}}}$$

$$\Psi_3 = 0$$

$$\Psi_4 = \frac{1}{4} \sqrt{\rho} \frac{\partial^2}{(\partial \rho)^2} f - \frac{\frac{\partial}{\partial \rho} f}{8 \sqrt{\rho}} + \frac{f}{8 \rho^{\frac{3}{2}}}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "II"